Closure of normal modal logics under the rule RK

The Open Logic Project contributors

openlogic:normal-modal-logic/axioms-systems:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-normal-modal-logic-axioms-systems/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the statement displays the RK inference figure via the prooftree environment, which the compatibility layer supplies from bussproofs-extra (present on the host); the proof is a complete induction carried in span

Editorial note. The label prop:rk names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Necessitation rule. Every normal modal logic is closed under the rule Necessitation: from A derive Box A.. The basis case n=1 of the induction is exactly Necessitation.
- Axiom K and modus ponens. Every normal modal logic contains all instances of Box(A implies B) implies (Box A implies Box B) and is closed under modus ponens and tautological reasoning.. The inductive step combines the induction hypothesis with the K instance for the final implication and closes by modus ponens with suitable tautological instances.

Proposition 0.1. *Every normal modal logic is closed under rule RK,*

$$\frac{A_1 \to (A_2 \to \cdots (A_{n-1} \to A_n) \cdots)}{\Box A_1 \to (\Box A_2 \to \cdots (\Box A_{n-1} \to \Box A_n) \cdots).}\ RK$$

Proof. By induction on n: If $n = 1$, then the rule is just NEC, and every normal modal logic is closed under NEC.

Now suppose the result holds for $n - 1$; we show it holds for n.

Assume

$$A_1 \to (A_2 \to \cdots (A_{n-1} \to A_n) \cdots) \in \Sigma$$

By the induction hypothesis, we have

$$\Box A_1 \to (\Box A_2 \to \cdots \Box(A_{n-1} \to A_n) \cdots) \in \Sigma$$

Since Σ is a normal modal logic, it contains all instances of K, in particular

$$\Box(A_{n-1} \to A_n) \to (\Box A_{n-1} \to \Box A_n) \in \Sigma$$

Using modus ponens and suitable tautological instances we get

$$\Box A_1 \to (\Box A_2 \to \cdots (\Box A_{n-1} \to \Box A_n) \cdots) \in \Sigma.$$

□